Pair objective with a monotonicity (inside-the-box) penalty

We optimize a pair of unit directions (d_1, d_2) jointly. Per anchor a and perturbation angle α, the norm-matched perturbations along d_1, d_2, and their equal mix give downstream L^2 responses $R_1(\alpha)$, $R_2(\alpha)$, $R_{\text{mix}}(\alpha)$.

Reward (unchanged). Push the mix into the plateau relative to the axes:

$$\mathcal{L}_{\text{ratio}} = \left\langle R_{\text{mix}}(\alpha) \,/\, (R_1(\alpha) + R_2(\alpha)) \right\rangle_{a,\alpha}.$$

Monotonicity penalty (new). The model $R = F\big(\sum_i w_i |\hat{f}_i \cdot v|^p\big)$ is a sum of non-negative terms, so adding a component cannot reduce the response: every iso-response point lies *at or inside the square*. Equivalently, the mix at α has the same projection onto d_1 as a pure-d_1 perturbation at α' with

$$\sin\alpha' = \sin\alpha \, \cos\tfrac{\pi}{4} = \sin\alpha/\sqrt{2},$$

plus an extra d_2 component, hence the model requires

$$R_{\text{mix}}(\alpha) \;\geq\; \max\big(R_1(\alpha'),\, R_2(\alpha')\big).$$

Cancellation (points *outside* the box) is the only way to violate this. Penalize the dimensionless shortfall:

$$\mathcal{L}_{\text{box}} = \left\langle \text{ReLU}\Big(1 - \frac{R_{\text{mix}}(\alpha)}{\max\big(R_1(\alpha'), R_2(\alpha')\big)}\Big) \right\rangle_{a,\alpha}.$$

It is 0 everywhere at/inside the box and positive only for cancellation.

Orthogonality (unchanged). $\mathcal{L}_{\text{orth}} = (\hat{d}_1 \cdot \hat{d}_2)^2$.

Total.

$$\mathcal{L} = \mathcal{L}_{\text{ratio}} + \lambda_{\text{box}}\,\mathcal{L}_{\text{box}} + \lambda_{\text{orth}}\,\mathcal{L}_{\text{orth}}.$$

Minimizing this still rewards a plateau-extending mix, but the optimizer can no longer reach $R_{\text{mix}} \approx 0$ by walking outside the model (cancellation); the lowest legitimate value sits at the box ($p \to \infty$).